Γ and X^2 Distributions

These two distributions are closely related and we'll discuss the specifics in a moment, but we'll start with some notation:

$\Gamma(\alpha, \beta) \rightarrow$ Gamma Distribution

$\chi^2(r) \rightarrow$ Chi-Squared distribution w/ r degrees of freedom

and that they both stem from the following function:

$$\int_0^\infty y^{\alpha-1} \cdot e^{-y}\, dy \quad \forall\, \alpha > 0$$

This integral is positive, and the function is known as the Gamma function of α, whereby:

$$\Gamma(\alpha) = \int_0^\infty y^{\alpha-1} \cdot e^{-y}\, dy = 1$$

if $\alpha = 1 \Rightarrow \Gamma(1) = \int_0^\infty e^{-y}\, dy = -e^{-y}\Big|_0^\infty = 0 - (-1) = 1$

if $\alpha > 1 \Rightarrow$ integration by parts

$$\Gamma(\alpha) = (\alpha-1)\int y^{\alpha-2} \cdot e^{-y}\, dy$$
$$= (\alpha-1) \cdot \Gamma(\alpha-1)$$

Thus, if α is a positive integer greater than 1, then:

$$\Gamma(\alpha) = (\alpha-1)\cdot(\alpha-2)\cdot \ldots \cdot (2)(1)\Gamma(1)$$
$$= (\alpha-1)!$$

since $\Gamma(1) = 1 \Rightarrow (1-1)! = 1$
$\Rightarrow 0! = 1$

Now, let y be a function of $x \ni y = \frac{x}{\beta}\ \forall\ \beta > 0$, then $dy = \frac{1}{\beta}dx$ and

$$\Gamma(\alpha) = \int \left(\frac{x}{\beta}\right)^{\alpha-1} e^{-(x/\beta)} \left(\frac{1}{\beta}\right) dx = 1$$

$$= \int_0^\infty \frac{1}{\beta^\alpha} x^{\alpha-1} e^{-\frac{x}{\beta}}\, dx = 1$$

$$\frac{\Gamma(\alpha)}{\Gamma(\alpha)} = 1 = \int_0^\infty \frac{1}{\Gamma(\alpha)\beta^\alpha} \cdot x^{\alpha-1} \cdot e^{-\frac{x}{\beta}}\, dx$$

so, assuming $\alpha > 0, \beta > 0$, and $\Gamma(\alpha) > 0$, then

$$f(x) = \begin{cases} \frac{1}{\Gamma(\alpha)\beta^\alpha} \cdot x^{\alpha-1} \cdot e^{-\frac{x}{\beta}}, & 0 < x < \infty \\ 0, & \text{o/w} \end{cases}$$

$$\Rightarrow x \sim \Gamma(\underbrace{\alpha}_{\text{shape}}, \underbrace{\beta}_{\text{scale}})$$

Interestingly, if we let $\alpha = 1$ and $\beta = \frac{1}{\lambda}$, then

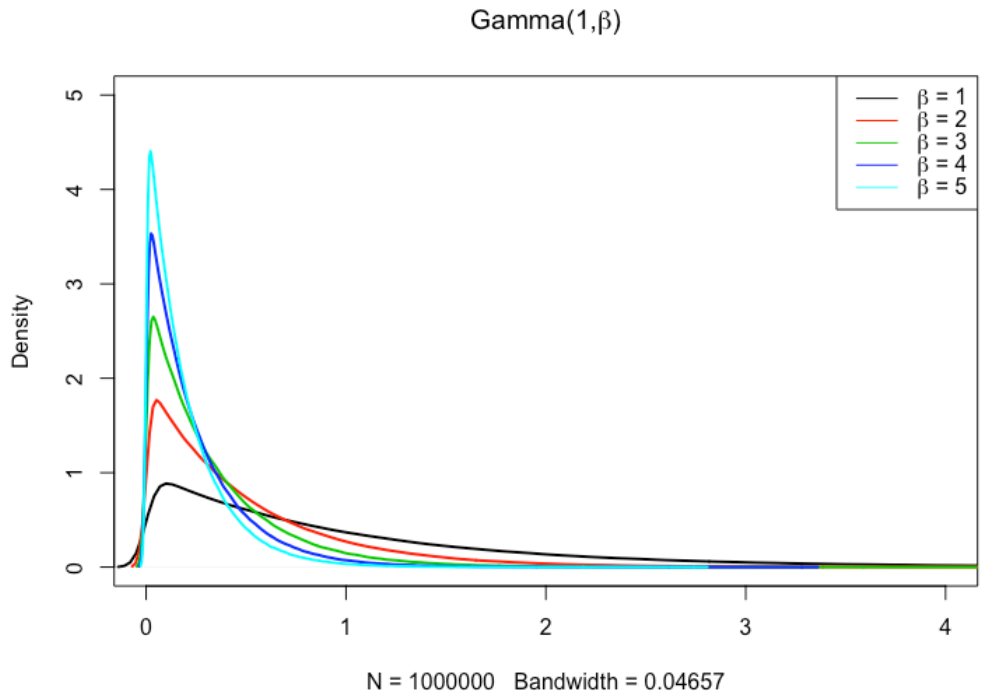
$$W \sim \Gamma\left(\alpha = 1, \beta = \frac{1}{\lambda}\right)$$

$$\Rightarrow f(w) = \frac{1}{\Gamma(1)(1/\lambda)^1} \cdot x^{1-1} \cdot e^{-\frac{x}{1/\lambda}}$$

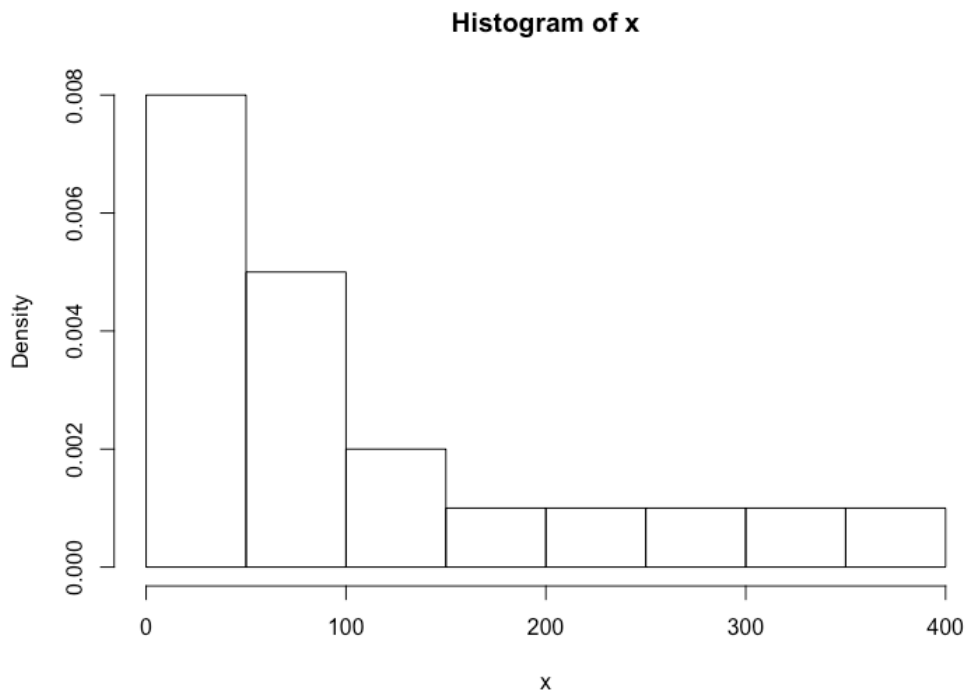
$$= \lambda e^{-\lambda x} \Rightarrow W \sim \exp(\lambda)$$

As such, we have a special case of the Gamma $\Gamma(\alpha, \beta)$ distribution that is known as the

exponential distribution




```
R Code:
> r <- 1000000
> y1 <- rgamma(r,1,1); d1 <- density(y1)
> y2 <- rgamma(r,1,2); d2 <- density(y2)
> y3 <- rgamma(r,1,3); d3 <- density(y3)
> y4 <- rgamma(r,1,4); d4 <- density(y4)
> y5 <- rgamma(r,1,5); d5 <- density(y5)
> yy <- c(0,5)
> xx <- c(0,4)
> tt <- expression(paste('Gamma(1,', beta,')',sep=''))
> ll <- c(expression(paste(beta,' = 1')),
          expression(paste(beta,' = 2')),
          expression(paste(beta,' = 3')),
          expression(paste(beta,' = 4')),
          expression(paste(beta,' = 5')))
> plot(d1, ylim=yy, xlim=xx, lwd=2, main=tt)
> lines(d2, ylim=yy, xlim=xx, lwd=2, col=2)
> lines(d3, ylim=yy, xlim=xx, lwd=2, col=3)
> lines(d4, ylim=yy, xlim=xx, lwd=2, col=4)
> lines(d5, ylim=yy, xlim=xx, lwd=2, col=5)
> legend('topright',ll,lty=1,col=1:5)
```




```
R Code:
> x <- c(1,4,5,21,22,28,40,42,51,53,58,67,
         95,124,124,160,202,260,303,363)
> hist(x,freq=F)
```

For another special case, let $Y = \alpha - 1$, wghere $\alpha \in \mathbb{Z}^+$

$$f(x) = \frac{1}{\Gamma(\alpha)\beta^{\alpha}} x^{\alpha-1} e^{-\frac{x}{\beta}}$$

$$= \frac{1}{(\alpha-1)!\,\beta^{\alpha}} x^{\alpha-1} e^{-\frac{x}{\beta}}$$

$$= \frac{1}{y!\,\beta^{\alpha}} x^{y} e^{-\frac{x}{\beta}}$$

Since $y = \alpha - 1 \Rightarrow \alpha = y+1$

$$\Rightarrow f(x) = \frac{1}{y!\,\beta^{y}\beta} x^{y} e^{-\frac{x}{\beta}}$$

$$= \frac{\left(\frac{x}{\beta}\right)^{y} e^{-\frac{x}{\beta}}}{y!}\left(\frac{1}{\beta}\right)$$

let $\frac{x}{\beta} = \lambda$

$$\Rightarrow f(x) = \underbrace{\frac{\lambda^{y} e^{-\lambda}}{y!}}_{\text{Functional form of the Poisson PMF}}\left(\frac{1}{\beta}\right)$$

if $X \sim \Gamma(\alpha, \beta)$ then $\forall x\ \exists\, \alpha \in \mathbb{Z}$ s.t. $P(X \le x) = P(Y \ge \alpha)$ where $Y \sim Pois\left(\lambda = \frac{x}{\beta}\right)$

The MGF of Gamma follows a similar procedure as before:

$$M(t) = E(e^{tX}) = \int_0^\infty e^{tx} \left[\frac{1}{\Gamma(\alpha)\cdot\beta^\alpha} x^{\alpha-1}\cdot e^{-\frac{x}{\beta}}\right] dx$$

and if we do a little bit of rearrangement

$$M(t) = \int_0^\infty \frac{1}{\Gamma(\alpha)\beta^\alpha} x^{\alpha-1} e^{-x(\frac{1}{\beta}-t)}\, dx$$

let $y = \left(\frac{1}{\beta}-t\right)x = \left(\frac{1-\beta t}{\beta}\right)x$

$\Rightarrow x = \left(\frac{\beta}{1-\beta t}\right)y$

$dx = \left(\frac{\beta}{1-\beta t}\right)dy$

$$M_y(t) = \int_0^\infty \frac{1}{\Gamma(\alpha)\beta^\alpha}\cdot\left[\left(\frac{\beta}{1-\beta t}\right)y\right]^{\alpha-1}\cdot e^{y}\cdot\left(\frac{\beta}{1-\beta t}\right)dy$$

$$= \int_0^\infty \frac{1}{\Gamma(\alpha)\beta^\alpha}\cdot\frac{\beta^{\alpha-1}}{(1-\beta t)^{\alpha-1}}\cdot y^{\alpha-1}\cdot\frac{\beta}{1-\beta t}\cdot e^{-y}\,dy$$

$$= \int_0^\infty \frac{1}{\Gamma(\alpha)}\cdot\frac{1}{(1-\beta t)^\alpha}\cdot y^{\alpha-1}\cdot e^{-y}\,dy$$

$$= \frac{1}{(1-\beta t)^\alpha}\underbrace{\int_0^\infty \frac{1}{\Gamma(\alpha)}\,y^{\alpha-1}e^{-y}\,dy}_{\Gamma(\alpha,\beta=1)}$$

β^α $\quad$ $1/\beta$

$$= \frac{1}{(1-\beta t)^\alpha}\cdot 1$$

valid pdf

$\Rightarrow M(t) = (1-\beta t)^{-\alpha}$ where $t < \frac{1}{\beta}$

Thus, we can now calculate our moments like normal by taking the appropriate derivative:

$$M'(t) = -\alpha(1-\beta t)^{-(\alpha+1)}(-\beta)$$
$$= \alpha\beta(1-\beta t)^{-(\alpha+1)}$$

$$M'(t=0) = \alpha\beta = E(x)$$

$$M''(t) = \alpha\beta(-(\alpha+1))(1-\beta t)^{-(\alpha+2)}(-\beta)$$
$$= \alpha\beta(\alpha+1)(\beta)(1-\beta t)^{-(\alpha+2)}$$
$$= \alpha(\alpha+1)\beta^2(1-\beta t)^{-(\alpha+2)}$$

$$M''(t=0) = \alpha(\alpha+1)\beta^2 = E(x^2)$$

$$Var(x) = E(x^2) - E(x)^2$$
$$= \alpha(\alpha+1)\beta^2 - \alpha^2\beta^2$$
$$= \alpha\beta^2$$

For another special case, let $\alpha = \frac{r}{2}$, where $r \in \mathbb{Z}^+$ and $\beta = 2$

$$\Longrightarrow \chi \sim \Gamma(\alpha = \tfrac{r}{2}, \beta = 2)$$

$$f(x) = \begin{cases} \frac{1}{\Gamma(\frac{r}{2})2^{r/2}} \cdot x^{\frac{r}{2}-1} e^{-\frac{x}{2}}, & 0 < x < \infty \\ 0, & e/w \end{cases}$$

The Chi-square distribution is used a lot in statistics for testing associations similar to the Pearson's correlation but when the two variables of interest are discrete random variables.

Real World Statistics: Chi-square distributions

Consider the relationship between smoking and lung cancer, we know that there is a direct causal relationship and, numerically, we would expect to see a difference between groups of smokers and non-smokers. Importantly, the count/frequency is not enough because, if we consider the two groups: smokers vs non-smokers, we know one is larger than the other. For a "fair" comparison, we look instead at the marginal relative frequencies/probabilities/rates to see if there is a difference between what we observe vs. what we would expect if the two variables are independent. A difference highlights that these two variables are not independent, which points towards an association.

$$X^2 = \frac{\sum(O_i - E_i)^2}{E_i}$$

Observed	D	D'
E	a	b
E'	c	d

Expected	D	D'
E	$\frac{(a+b)(a+c)}{n}$	$\frac{(a+b)(b+d)}{n}$
E'	$\frac{(a+c)(c+d)}{n}$	$\frac{(b+d)(c+d)}{n}$

(To be continued)

$X \sim \Gamma(\alpha, \beta) \Rightarrow M(t) = (1-\beta t)^{-\alpha}$

$X \sim \chi^2 \equiv \Gamma(\alpha = \frac{r}{2}, \beta = 2)$

$\Rightarrow M(t) = (1-2t)^{-\frac{r}{2}}, \quad t < \frac{1}{2}$

$\therefore \quad E(x) = \alpha\beta = (\frac{r}{2})(2) = r$

$Var(x) = \alpha\beta^2 = (\frac{r}{2})(2)^2 = 2r$

Notation:

$X \sim \chi^2(r)$ or $X \sim \chi^2_r$

Example 3.7: Let $X \sim f(x) \ni f(x) = \frac{1}{4}x \cdot e^{-\frac{x}{2}}$

$f(x) = \frac{1}{4}x^1 \cdot e^{-\frac{x}{2}}$

Recall: $\Gamma(\alpha, \beta)$, the exponential involves $e^{-\frac{x}{\beta}}$

$\therefore \beta = 2 \Rightarrow X \sim \chi^2$

$\Gamma(\alpha, \beta)$ has $x^{\alpha-1} \Rightarrow \alpha = 2$

$\Rightarrow \Gamma(2, 2)$

$X \sim \chi^2(4)$

$\frac{r}{2} = 2 \Rightarrow r = 4$

Theorem 3.3.1: Let $X \sim \chi^2(r)$, if $k > -\frac{r}{2}$ then $E(X^k)$ $\exists$ and is given by:

$$E(X^k) = \frac{2^k \cdot \Gamma\left(\frac{r}{2} + k\right)}{\Gamma\left(\frac{r}{2}\right)}$$

That is, we have the general formula for the kth moment. This should not be too surprising, as we've seen before such recursive function.

Example 3.8: If $X \sim Exp(\lambda)$, then

$$\left.\begin{aligned} M(t) &= \lambda(\lambda - t)^{-1} \\ M'(t) &= \lambda(\lambda - t)^{-2} \\ M''(t) &= 2\lambda(\lambda - t)^{-3} \\ M^{(3)}(t) &= 6\lambda(\lambda - t)^{-4} \end{aligned}\right\} M^{(k)}(t) = k!\,\lambda(\lambda - t)^{-(k+1)}$$

So, if we let $X \sim \Gamma\left(\alpha = \frac{r}{2}, \beta = 2\right)$ then $M(t) = (1 - 2t)^{-\frac{r}{2}}$ and our moments are:

$$M(t) = (1-2t)^{-\frac{r}{2}}$$

$$M'(t) = \left(-\frac{r}{2}\right)(1-2t)^{-\left(\frac{r}{2}+1\right)}(-2) = 2\left(\frac{r}{2}\right)(1-2t)^{-\left(\frac{r}{2}+1\right)}$$

$$M''(t) = 2^2\left(\frac{r}{2}\right)\left(\frac{r}{2}+1\right)(1-2t)^{-\left(\frac{r}{2}+2\right)}$$

$$M^{(3)}(t) = 2^3\left(\frac{r}{2}\right)\left(\frac{r}{2}+1\right)\left(\frac{r}{2}+2\right)(1-2t)^{-\left(\frac{r}{2}+3\right)}$$

$$= \frac{2^3\left(\frac{r}{2}\right)\left(\frac{r}{2}+1\right)\left(\frac{r}{2}+2\right)(1-2t)^{-\left(\frac{r}{2}+3\right)}\left(\frac{r}{2}-1\right)!}{\left(\frac{r}{2}-1\right)!} \quad \text{Note } \Gamma(\alpha) = (\alpha-1)!$$

$$= \frac{2^3\left(\frac{r}{2}+2\right)!(1-2t)^{-\left(\frac{r}{2}+3\right)}}{\left(\frac{r}{2}+1\right)!} = \frac{2^3\Gamma\left(\frac{r}{2}+3\right)(1-2t)^{-\left(\frac{r}{2}+3\right)}}{\Gamma\left(\frac{r}{2}\right)}$$

$$M^{(k)}(t) = \frac{2^k\Gamma\left(\frac{r}{2}+k\right)(1-2t)^{-\left(\frac{r}{2}+k\right)}}{\Gamma\left(\frac{r}{2}\right)}$$

$$M^{(k)}(t) = 2^k\left[\prod_{i=0}^{k-1}\left(\frac{r}{2}+i\right)\right](1-2t)^{-\left(\frac{r}{2}+k\right)}$$

$$M^{(k)}(t=0) = \frac{2^k\Gamma\left(\frac{r}{2}+k\right)}{\Gamma\left(\frac{r}{2}\right)}$$

Theorem 3.3.2: Let $X_1, X_2, \dots, X_n \overset{iid}{\sim} \Gamma(\alpha_i, \beta)$, then if $Y = \sum X_i$, $Y \sim \Gamma(\sum \alpha_i, \beta)$

Corollary 3.3.1: Let $X_1, X_2, \dots, X_n \overset{iid}{\sim} \Gamma(\alpha_i, \beta = 2) \ni \alpha_i = \frac{r_i}{2}\ \forall\, r_i \in \mathbb{Z}^+$, then if $Y = \sum X_i$, $Y \sim \mathrm{X}^2(\sum r_i)$

Example 3.9: Let $X \sim f(x) \ni M(t) = (1-2t)^{-6}$, calculate $P(X < 5.23)$

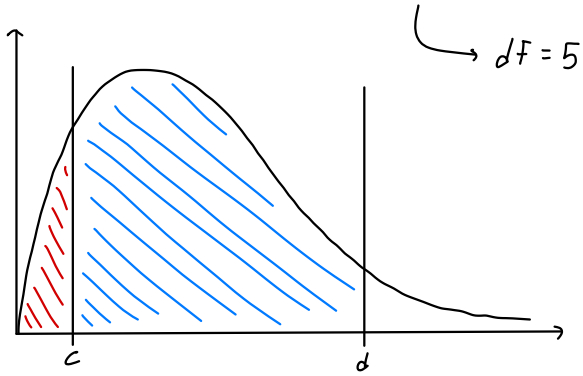
$X \sim \Gamma(\alpha = 6, \beta = 2) \Rightarrow X \sim \chi^2(r = 12)$

$P(X < 5.23 \mid df = r = 12) = 0.05$

$\hookrightarrow 0.05014788$

Example 3.10: Let $X \sim \mathrm{X}^2(r = 5)$, find $c, d \ni P(c < X < d) = 0.95$ and $P(X < c) = 0.025$.

$\hookrightarrow df = 5$



$P(X < c) = 0.025 \Rightarrow c = 0.831$

i.e. $P(X < 0.831) = 0.025$

$$\begin{aligned} P(c < X < d) &= P(c < X \cap X < d) \\ &= P(X < d) + P(X > c) - P(X < d \cup X > c) \\ &= P(X < d) + (1 - P(X < c)) - 1 \\ &= P(X < d) - P(X \le c) \end{aligned}$$

$\Rightarrow P(X < d) = 0.95 + 0.025$

$P(X < d) = 0.975$

$(r = 5\ df)$

$d = 12.833$

$\therefore Pr(X < 12.833 \mid r = 5) = 0.975$

As mentioned previously, there are close ties between many of the previously established distributions from STAT101/231 and the $\Gamma(\alpha, \beta)$, with the newer $\mathrm{X}^2(r)$ being the most obvious. We've also seen how a random variable following the $\mathrm{X}^2(r)$ distribution is a little easier to work with since we have a table to use. Thus, if we have $X \sim \Gamma(\alpha, \beta)$, we can answer any problem using $Y \sim \mathrm{X}^2(r)$ by performing a transformation.

Example 3.11:

let $X \sim \Gamma(\alpha = 3, \beta = 4)$ what is $P(3.28 < X < 25.2)$

let $Y = \frac{1}{2}X \Rightarrow P(1.64 < Y < 12.6)$

$X \sim f(x) = \frac{1}{\Gamma(3)4^3} x^2 e^{-\frac{x}{4}}$

$Y = \frac{1}{2}X \Rightarrow X = 2Y$

$\Rightarrow f_Y(y) = f_X(g^{-1}(y)) \left| \frac{d}{dy} x \right|$

$= \frac{1}{\Gamma(3)4^3}(2y)^2 e^{-\frac{2y}{4}} |2|$

$= \frac{1}{\Gamma(3)2^6} \cdot 2^2 y^2 \cdot e^{-\frac{y}{2}} \cdot 2$

$= \frac{1}{\Gamma(3)2^3} y^2 e^{-\frac{y}{2}}$

$Y \sim \Gamma(\alpha = 3, \beta = 2)$

$Y \sim \chi^2(r = 6)$

$P(1.64 < Y < 12.6 \mid r = 6) = 0.95 - 0.05 = 0.90$

Table II
Chi-Square Distribution

The following table presents selected quantiles of chi-square distribution, i.e., the values x such that

$$P(X \leq x) = \int_0^x \frac{1}{\Gamma(r/2)2^{r/2}} w^{r/2-1} e^{-w/2}\, dw,$$

for selected degrees of freedom r.

	$P(X \leq x)$							
r	0.010	0.025	0.050	0.100	0.900	0.950	0.975	0.990
1	0.000	0.001	0.004	0.016	2.706	3.841	5.024	6.635
2	0.020	0.051	0.103	0.211	4.605	5.991	7.378	9.210
3	0.115	0.216	0.352	0.584	6.251	7.815	9.348	11.345
4	0.297	0.484	0.711	1.064	7.779	9.488	11.143	13.277
5	0.554	0.831	1.145	1.610	9.236	11.070	12.833	15.086
6	0.872	1.237	1.635	2.204	10.645	12.592	14.449	16.812
7	1.239	1.690	2.167	2.833	12.017	14.067	16.013	18.475
8	1.646	2.180	2.733	3.490	13.362	15.507	17.535	20.090
9	2.088	2.700	3.325	4.168	14.684	16.919	19.023	21.666
10	2.558	3.247	3.940	4.865	15.987	18.307	20.483	23.209
11	3.053	3.816	4.575	5.578	17.275	19.675	21.920	24.725
12	3.571	4.404	5.226	6.304	18.549	21.026	23.337	26.217
13	4.107	5.009	5.892	7.042	19.812	22.362	24.736	27.688
14	4.660	5.629	6.571	7.790	21.064	23.685	26.119	29.141
15	5.229	6.262	7.261	8.547	22.307	24.996	27.488	30.578
16	5.812	6.908	7.962	9.312	23.542	26.296	28.845	32.000
17	6.408	7.564	8.672	10.085	24.769	27.587	30.191	33.409
18	7.015	8.231	9.390	10.865	25.989	28.869	31.526	34.805
19	7.633	8.907	10.117	11.651	27.204	30.144	32.852	36.191
20	8.260	9.591	10.851	12.443	28.412	31.410	34.170	37.566
21	8.897	10.283	11.591	13.240	29.615	32.671	35.479	38.932
22	9.542	10.982	12.338	14.041	30.813	33.924	36.781	40.289
23	10.196	11.689	13.091	14.848	32.007	35.172	38.076	41.638
24	10.856	12.401	13.848	15.659	33.196	36.415	39.364	42.980
25	11.524	13.120	14.611	16.473	34.382	37.652	40.646	44.314
26	12.198	13.844	15.379	17.292	35.563	38.885	41.923	45.642
27	12.879	14.573	16.151	18.114	36.741	40.113	43.195	46.963
28	13.565	15.308	16.928	18.939	37.916	41.337	44.461	48.278
29	14.256	16.047	17.708	19.768	39.087	42.557	45.722	49.588
30	14.953	16.791	18.493	20.599	40.256	43.773	46.979	50.892

Normal Distribution

The standard normal distribution obviously plays an important role in many statistical inferences; however, we have seen this distribution many times so we will not dwell on it and focus on the newer ideas, starting with the moment generating functions. Let $Z \sim N(0,1)$, then

$$M(t) = E(e^{tZ}) = \int e^{tZ} \cdot \frac{1}{\sqrt{2\pi}} e^{-\frac{z^2}{2}} dz$$

$$= \int \frac{1}{\sqrt{2\pi}} e^{-\frac{z^2}{2} + \frac{2tz}{2}} dz$$

$$= \int \frac{1}{\sqrt{2\pi}} e^{-\frac{1}{2}(z^2 - 2tz)} dz$$

$$= \int \frac{1}{\sqrt{2\pi}} e^{-\frac{1}{2}(z^2 - 2tz + t^2)} e^{\frac{1}{2}t^2} dz$$

$$= e^{\frac{1}{2}t^2} \int \frac{1}{\sqrt{2\pi}} e^{-\frac{1}{2}(z-t)^2} dz$$

let $w = z - t$

$\Rightarrow dw = dz$

$$= e^{\frac{1}{2}t^2} \underbrace{\int \frac{1}{\sqrt{2\pi}} e^{-\frac{1}{2}w^2} dw}_{\text{PDF for } W \sim N(0,1),\ \int \cdots dw = 1}$$

$$= e^{\frac{1}{2}t^2} \cdot 1$$

$$= e^{\frac{1}{2}t^2}$$

$M(t) = e^{\frac{1}{2}t^2}$

$M'(t) = t e^{\frac{1}{2}t^2}$

$M''(t) = e^{\frac{1}{2}t^2} + t^2 e^{\frac{1}{2}t^2}$

$M'(t=0) = 0$

$M''(t=0) = 1$

$\Rightarrow \mu = 0$

$\sigma^2 = 1^2 - 0^2 = 1$

If $X_1, X_2, \dots, X_n \overset{iid}{\sim} \mathrm{N}(0,1)$ and $Y = \sum X_i^2$, then using the same MGF methods we've used before when dealing with IND random variables, we can show that:

$$M_Y(t) = E(e^{tY}) = E\left(e^{t[X_1^2 + X_2^2 + \dots + X_n^2]}\right)$$
$$= E(e^{tX_1^2}) \cdot E(e^{tX_2^2}) \cdot \dots \cdot E(e^{tX_n^2})$$
$$= \prod_{i=1}^{n} E(e^{tX_i^2})$$

For any generic i, we know that $E\left(e^{tX_i^2}\right) \equiv E\left(e^{tZ_i^2}\right)$, which is equal to:

$$E(e^{tZ_i^2}) = \int e^{tZ_i^2} \cdot \frac{1}{\sqrt{2\pi}} e^{-\frac{1}{2}Z_i^2} \, dZ_i$$
$$= \int \frac{1}{\sqrt{2\pi}} e^{tZ_i^2 - \frac{1}{2}Z_i^2} \, dZ_i$$
$$= \int \frac{1}{\sqrt{2\pi}} e^{\frac{2tZ_i^2}{2} - \frac{1}{2}Z_i^2} \, dZ_i$$
$$= \int \frac{1}{\sqrt{2\pi}} e^{-\frac{1}{2}Z_i^2(1-2t)} \, dZ_i$$
$$= \int \frac{1}{\sqrt{2\pi}} e^{-\frac{1}{2}w^2} \cdot (1-2t)^{-\frac{1}{2}} \, dw \qquad \text{let } w = z \cdot \sqrt{1-2t}$$
$$= (1-2t)^{-\frac{1}{2}} \int \frac{1}{\sqrt{2\pi}} e^{-\frac{1}{2}w^2} \, dw \qquad \Rightarrow dw = \sqrt{1-2t}\, dz$$
$$= (1-2t)^{\frac{1}{2}} dz$$
$$= (1-2t)^{-\frac{1}{2}}$$

$$M_Y(t) = \prod_{i=1}^{n} E(e^{tZ_i^2})$$
$$= \prod_{i=1}^{n} (1-2t)^{-\frac{1}{2}}$$
$$= \left[(1-2t)^{-\frac{1}{2}}\right]^n$$
$$= (1-2t)^{-\frac{n}{2}}$$
$$\Rightarrow Y = \sum X_i^2 \equiv \sum_{i=1}^{n} Z_i^2 \sim \chi^2(r=n)$$

That is to say, the sum of the square of independent random variables that follow a standard normal distribution will follow a $\mathrm{X}^2(r)$ distribution, which links the normal distribution back to our $\Gamma(\alpha, \beta)$ distribution.

Real World Statistics: Chi-square distributions (continued)

Let's consider the following 2x2 table of frequencies relating to D-E status:

		D	D'	
exposed →	E	20	20	40
Healthy →	E'	10	50	60
		30	70	100

The frequency of disease, which can be thought of as the probability of disease, is 30/100 = 0.30. If we wanted to, we could compare two rates (D|E) vs (D|E') and perform a two-independent proportion test like in introductory statistics. Alternatively, we could use the X^2 as previously described.

If there was no association between E and D, then the frequency of D should be the same between the groups (i.e. we shouldn't expect a difference). As such, if there was no association then we could say that E and D are independent, so:

$$P(D|E) = P(D) = P(D|E')$$

Therefore, if we look at the E group, we would expect the D count to be:

$$40\left(\frac{30}{100}\right) = 12$$

and that among the E' group, we would expect the D count to be:

$$60\left(\frac{30}{100}\right) = 18$$

We can do the same for the expected frequency based on D' to fill in the other column. Thus, our expected frequency table would look like:

	D	D'	
E	12	28	40
E'	18	42	60
	30	70	100

This tells us that if E and D were independent, this is the expected frequency and it clearly looks very different from what we observed.

$$\chi^2_{cal} = \sum\sum \frac{(O_{ij}-E_{ij})^2}{E_{ij}} \sim \chi^2\big(r = [row-1]\cdot[col-1]\big)$$

$r=1$ for 2×2 tables

$$= 12.698$$

test statistic

$$\text{p-value} = P(\chi^2 > 12.698 \mid r=1) <<< \alpha = 0.01$$

$$12.698 >> 6.635$$

*need complement

Thus, we would conclude that there is evidence (really strong evidence, the actual p-value = 0.000366) that there is an association between E and D.

Previously, we talked about the MGF for $Z \sim N(0,1)$. Let $X = bZ + a$, i.e. X is a linear combination of Z.

$$\Rightarrow Z = \frac{1}{b}(X-a)$$

$$dz = \frac{1}{b}x$$

?

$$\therefore J = \frac{d}{dz}X = \frac{1}{b}$$

$$\Rightarrow f_x(x) = f_z(g^{-1}(x))\cdot|J|$$

$$= f_z\left(\frac{1}{b}(x-a)\right)\cdot|J|$$

$$= \frac{1}{\sqrt{2\pi}} e^{-\frac{1}{2}\left(\frac{x-a}{b}\right)^2} \cdot \frac{1}{b}$$

$$= \frac{1}{b\sqrt{2\pi}} e^{-\frac{1}{2}\left(\frac{x-a}{b}\right)^2}$$

$$E(x) = E(bz+a) = bE(z)+a = b(0)+a = a = \mu_x$$

$$Var(x) = Var(bz+a) = b^2 Var(z) = b^2(1) = b^2 = \sigma_x^2$$

$$\Rightarrow f(x) = \frac{1}{\sqrt{2\pi}\,\sigma_x} e^{-\frac{1}{2}\left(\frac{x-\mu_x}{\sigma_x}\right)^2}$$

$\therefore X \sim N(\mu, \sigma)$

For the MGF of X, since $X = bZ + a = \sigma \cdot Z + \mu$,

$$E(e^{tX}) = E(e^{t(\sigma Z + \mu)})$$

$$= E(e^{t\sigma Z + \mu t})$$

$$= E(e^{t\sigma Z} \cdot e^{\mu t})$$

$$= E(e^{t\sigma Z}) \cdot E(e^{\mu t})$$

$$= e^{\mu t} \cdot E(e^{t\sigma Z})$$

$$= e^{\mu t} \int e^{t\sigma z} \cdot \frac{1}{\sqrt{2\pi}} e^{-\frac{1}{2} z^2} dz$$

$\vdots$ [we did this before by completing the square]

$$= e^{\mu t} \cdot e^{\frac{1}{2}\sigma^2 t^2} \int \frac{1}{\sqrt{2\pi}} e^{-\frac{1}{2}(z - \sigma t)^2} dx$$

$$= e^{\mu t} \cdot e^{\frac{1}{2}\sigma^2 t^2} \cdot 1$$

$$M(t) = e^{\mu t + \frac{1}{2}\sigma^2 t^2}$$

$$M'(t) = e^{\mu t + \frac{1}{2}\sigma^2 t^2} (\mu + \sigma^2 t)$$

$$M'(t=0) = \mu$$

$$M''(t) = e^{\mu t + \frac{1}{2}\sigma^2 t^2}(\mu + \sigma^2 t)^2 + e^{\mu t + \frac{1}{2}\sigma^2 t^2}(\sigma^2)$$

$$M''(t=0) = \mu^2 + \sigma^2$$

$$\begin{aligned} Var(X) &= E(X^2) - E(X)^2 \\ &= \mu^2 + \sigma^2 - \mu^2 \\ &= \sigma^2 \end{aligned}$$

Example 3.12:

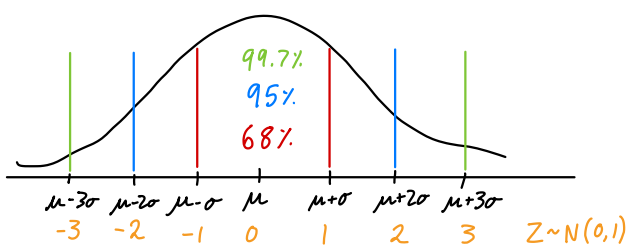
$$M(t) = e^{2t + 32t^2} = e^{2t(1+16t)}$$

$$X \sim N(\mu = 2, \sigma = 8)$$

Example 3.13:

Suppose $X \sim N(\mu, \sigma)$ s.t. $Pr(X < 60) = 0.10$

$Pr(X > 90) = 0.05$



$$Z = \frac{X-\mu}{\sigma} \qquad P(Z < c) = p\%$$

$$\frac{60-\mu}{\sigma} \qquad P(Z<c) = 0.10 \Rightarrow c = \Phi^{-1}(0.10)$$

using our tables, $c = -1.282$

$$\frac{90-\mu}{\sigma} = \quad P(Z<d) = 1 - 0.05 = 0.95 \Rightarrow d = \Phi^{-1}(0.95)$$

$$d = 1.645$$

$$\left.\begin{aligned} \frac{60-\mu}{\sigma} &= -1.282 \\ \frac{90-\mu}{\sigma} &= 1.645 \end{aligned}\right\} \begin{aligned} \mu &= 73.138 \\ \sigma &= 10.251 \end{aligned}$$

Theorem 3.4.1: If the random variable $X \sim N(\mu, \sigma) \ \ni \ \sigma > 0$, then the random variable

$$V = \left(\frac{X-\mu}{\sigma}\right)^2 \sim \mathrm{X}^2(r=1)$$

z

Theorem 3.4.2: If $X_1, X_2, \ldots, X_n \overset{iid}{\sim} \mathrm{N}(\mu_i, \sigma_i)$ and $Y = \sum a_i \cdot X_i$, where $a_i \in \mathbb{R}$, then

$$Y \sim N\left(\sum a_i \cdot \mu_i, \sqrt{\sum a_i^2 \cdot \sigma_i^2}\right)$$

An important an special case of this theorem is when $Y = \frac{1}{n}\sum X_i$, $a_i = \frac{1}{n}$ $\forall\, i$ and $X_i \sim N(\mu, \sigma)$

Corollary 3.4.1: If $X_1, X_2, \ldots, X_n \overset{iid}{\sim} \mathrm{N}(\mu, \sigma)$ and $Y = \frac{1}{n}\sum X_i$, then

$$Y \sim N\left(\mu, \frac{\sigma}{\sqrt{n}}\right)$$

$X \overset{iid}{\sim} N(\mu, \sigma)$, $a_i = \frac{1}{n}$

$\Rightarrow Y = \frac{1}{n}\sum x_i = \bar{X} = \left(\frac{1}{n}x_1 + \ldots + \frac{1}{n}x_2\right)$

$\Rightarrow \sum a_i \mu_i = \sum a_i \mu = \sum \frac{1}{n}\mu = n\left(\frac{1}{n}\cdot\mu\right) = \mu$

$E(X) = E\left(\frac{1}{n}\sum X_i\right) = \frac{1}{n}E\left(\sum X_i\right) = \frac{1}{n}\sum E(X_i) = \frac{1}{n}[n\mu] = \mu$

$$\begin{aligned}
Var(Y) = Var\left(\tfrac{1}{n}X_1 + \ldots + \tfrac{1}{n}X_n\right) &= Var\left(\tfrac{1}{n}X_1\right) + \ldots + Var\left(\tfrac{1}{n}X_n\right) \\
&= \tfrac{1}{n^2}Var(X_1) + \ldots + \tfrac{1}{n^2}Var(X_n) \\
&= n\left(\tfrac{1}{n^2}\cdot\sigma^2\right) \\
&= \frac{\sigma^2}{n} \\
&\Rightarrow \sigma_{\bar{X}} = \sqrt{\frac{\sigma^2}{n}} = \frac{\sigma}{\sqrt{n}}
\end{aligned}$$

Importantly, we can see that this theorem and corollary is a special case when $Y = \frac{1}{n}\sum X_i = \bar{X}$, and is the basis for the **Central Limit Theorem**.

Table III
Normal Distribution

The following table presents the standard normal distribution. The probabilities tabled are

$$P(Z \leq z) = \Phi(z) = \int_{-\infty}^{z} \frac{1}{\sqrt{2\pi}} e^{-w^2/2}\, dw.$$

Note that only the probabilities for $z \geq 0$ are tabled. To obtain the probabilities for $z < 0$, use the identity $\Phi(-z) = 1 - \Phi(z)$. At the bottom of the table, some useful quantiles of the standard normal distribution are displayed.

z	0.00	0.01	0.02	0.03	0.04	0.05	0.06	0.07	0.08	0.09
0.0	.5000	.5040	.5080	.5120	.5160	.5199	.5239	.5279	.5319	.5359
0.1	.5398	.5438	.5478	.5517	.5557	.5596	.5636	.5675	.5714	.5753
0.2	.5793	.5832	.5871	.5910	.5948	.5987	.6026	.6064	.6103	.6141
0.3	.6179	.6217	.6255	.6293	.6331	.6368	.6406	.6443	.6480	.6517
0.4	.6554	.6591	.6628	.6664	.6700	.6736	.6772	.6808	.6844	.6879
0.5	.6915	.6950	.6985	.7019	.7054	.7088	.7123	.7157	.7190	.7224
0.6	.7257	.7291	.7324	.7357	.7389	.7422	.7454	.7486	.7517	.7549
0.7	.7580	.7611	.7642	.7673	.7704	.7734	.7764	.7794	.7823	.7852
0.8	.7881	.7910	.7939	.7967	.7995	.8023	.8051	.8078	.8106	.8133
0.9	.8159	.8186	.8212	.8238	.8264	.8289	.8315	.8340	.8365	.8389
1.0	.8413	.8438	.8461	.8485	.8508	.8531	.8554	.8577	.8599	.8621
1.1	.8643	.8665	.8686	.8708	.8729	.8749	.8770	.8790	.8810	.8830
1.2	.8849	.8869	.8888	.8907	.8925	.8944	.8962	.8980	.8997	.9015
1.3	.9032	.9049	.9066	.9082	.9099	.9115	.9131	.9147	.9162	.9177
1.4	.9192	.9207	.9222	.9236	.9251	.9265	.9279	.9292	.9306	.9319
1.5	.9332	.9345	.9357	.9370	.9382	.9394	.9406	.9418	.9429	.9441
1.6	.9452	.9463	.9474	.9484	.9495	.9505	.9515	.9525	.9535	.9545
1.7	.9554	.9564	.9573	.9582	.9591	.9599	.9608	.9616	.9625	.9633
1.8	.9641	.9649	.9656	.9664	.9671	.9678	.9686	.9693	.9699	.9706
1.9	.9713	.9719	.9726	.9732	.9738	.9744	.9750	.9756	.9761	.9767
2.0	.9772	.9778	.9783	.9788	.9793	.9798	.9803	.9808	.9812	.9817
2.1	.9821	.9826	.9830	.9834	.9838	.9842	.9846	.9850	.9854	.9857
2.2	.9861	.9864	.9868	.9871	.9875	.9878	.9881	.9884	.9887	.9890
2.3	.9893	.9896	.9898	.9901	.9904	.9906	.9909	.9911	.9913	.9916
2.4	.9918	.9920	.9922	.9925	.9927	.9929	.9931	.9932	.9934	.9936
2.5	.9938	.9940	.9941	.9943	.9945	.9946	.9948	.9949	.9951	.9952
2.6	.9953	.9955	.9956	.9957	.9959	.9960	.9961	.9962	.9963	.9964
2.7	.9965	.9966	.9967	.9968	.9969	.9970	.9971	.9972	.9973	.9974
2.8	.9974	.9975	.9976	.9977	.9977	.9978	.9979	.9979	.9980	.9981
2.9	.9981	.9982	.9982	.9983	.9984	.9984	.9985	.9985	.9986	.9986
3.0	.9987	.9987	.9987	.9988	.9988	.9989	.9989	.9989	.9990	.9990
3.1	.9990	.9991	.9991	.9991	.9992	.9992	.9992	.9992	.9993	.9993
3.2	.9993	.9993	.9994	.9994	.9994	.9994	.9994	.9995	.9995	.9995
3.3	.9995	.9995	.9995	.9996	.9996	.9996	.9996	.9996	.9996	.9997
3.4	.9997	.9997	.9997	.9997	.9997	.9997	.9997	.9997	.9997	.9998
3.5	.9998	.9998	.9998	.9998	.9998	.9998	.9998	.9998	.9998	.9998

α	0.400	0.300	0.200	0.100	0.050	0.025	0.020	0.010	0.005	0.001
z_α	0.253	0.524	0.842	1.282	1.645	1.960	2.054	2.326	2.576	3.090
$z_{\alpha/2}$	0.842	1.036	1.282	1.645	1.960	2.241	2.326	2.576	2.807	3.291